Avishkar Solar Sync

- *AKSSHIT AARYA ABHINAV*
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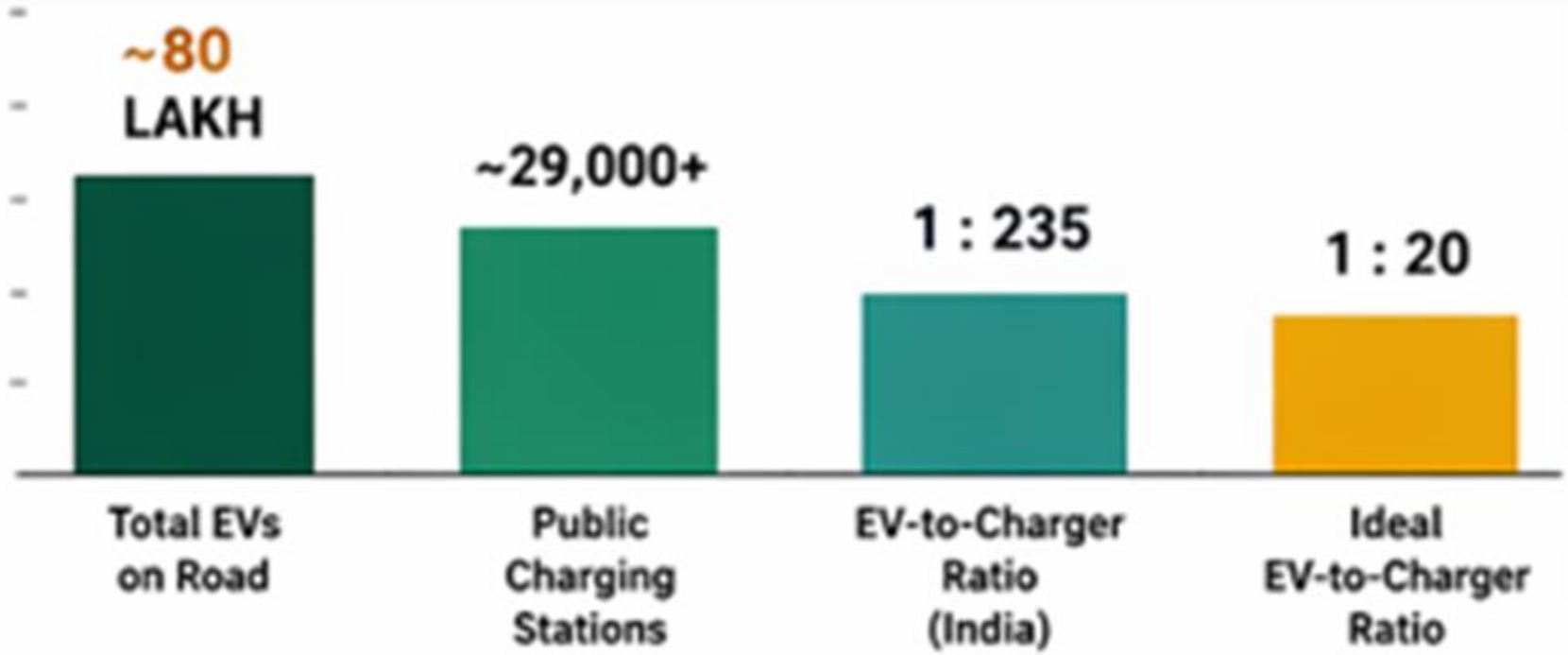
PROBLEM STATEMENT

The rapid adoption of electric scooters in India is limited by **inadequate charging infrastructure, long charging times, and range anxiety among users**. Many users, especially students and daily commuters, do not have easy or reliable access to charging points when they are outside their homes.

At the same time, India receives **abundant sunlight** throughout the year, yet this renewable energy source remains largely **untapped in personal transportation**.

Category	Numbers (India)	Insight
Total EVs on Road	8 million	Rapid growth in last 3 – 4 years
Annual EV Sales	23+ lakh (2025)	8% of all vehicle sales
Electric Scooters (2W share)	80%+ of EV sales	Dominates Indian EV market
Public Charging Stations	**~29,000+**	**Growing but still insufficient**
EV-to-Charger Ratio	1 : 235	Ideal is 1 : 20
Fast Chargers	**8,800**	**Limited availability**
Slow Chargers	20,000+	Majority of network



EV adoption is growing rapidly, but charging infrastructure is far behind.

CURRENT GAPS & COMPETITION ANALYSIS

1. Charging Infrastructure Gap

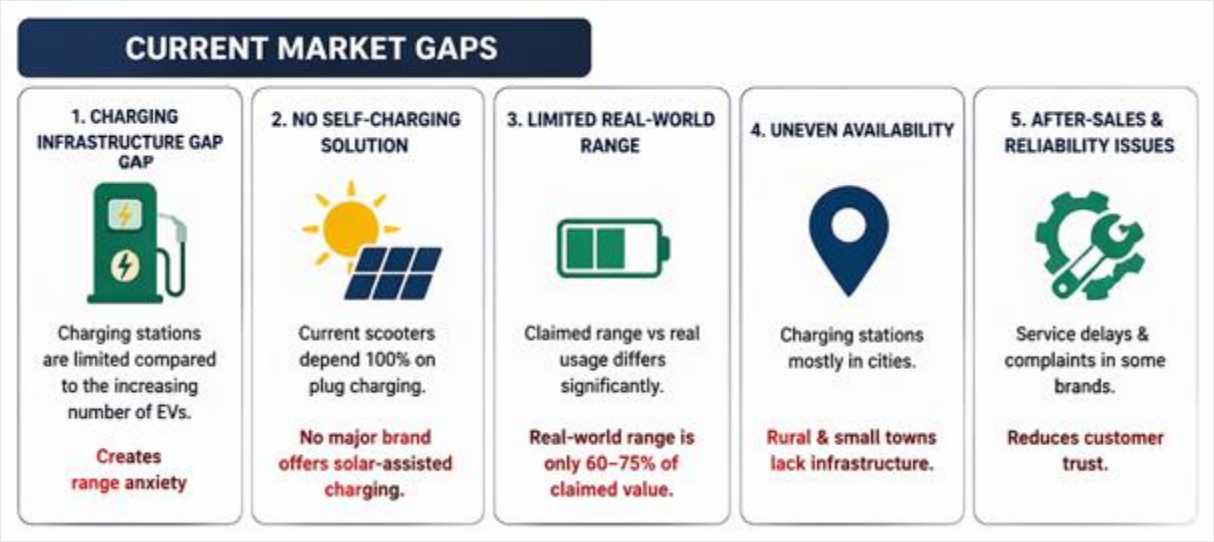
EV adoption is growing rapidly, but charging stations are limited **EV-to-charger ratio** is very high. Which creates range anxiety.

2. No Self-Charging Solution

All current scooters depend **100% on plug charging**

3. Limited Real-World range

Ev company claims the range that are tested in 'Controlled environment'. Real range in city reduces to 60% - 70% of claimed value.



CURRENT GAPS & COMPETITION ANALYSIS

SOLARSYNC E-SCOOTER

CURRENT MARKET GAPS

1. CHARGING INFRASTRUCTURE GAP GAP

Charging stations are limited compared to the increasing number of EVs.

Creates range anxiety

2. NO SELF-CHARGING SOLUTION

Current scooters depend 100% on plug charging.

No major brand offers solar-assisted charging.

3. LIMITED REAL-WORLD RANGE

Claimed range vs real usage differs significantly.

Real-world range is only 60–75% of claimed value.

4. UNEVEN AVAILABILITY

Charging stations mostly in cities.

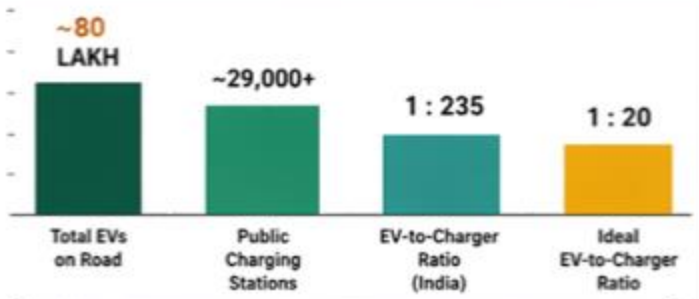
Rural & small towns lack infrastructure.

5. AFTER-SALES & RELIABILITY ISSUES

Service delays & complaints in some brands.

Reduces customer trust.

INDIA EV ECOSYSTEM OVERVIEW (2025)



EV adoption is growing rapidly, but charging infrastructure is far behind.

COMPETITION ANALYSIS – MAJOR PLAYERS

TVS iQUBE

- Market leader
- Strong dealer network
- Good reliability

BAJAJ CHETAK

- Trusted brand
- Strong build quality
- Premium positioning

ATHER ENERGY

- Advanced technology
- Premium features
- High performance

OLA ELECTRIC

- Price aggressive
- Wide product range
- Service challanges

HERO VIDA

- Affordable
- Growing fast
- Good features

COMPETITOR WEAKNESS vs OUR ADVANTAGE

FACTOR	CURRENT SCOOTERS	OUR SOLUTION ADVANTAGE
Charging	Only plug-based charging	Solar + Plug charging (dual advantage)
Range	Limited to battery capacity	Extra range boost from solar energy
Infrastructure Dependency	High dependency on charging stations	Reduced dependency, charge anywhere with sun
Innovation	Incremental upgrades in features	Unique solar-assisted self-charging solution

OUR EDGE

While others improve performance, we solve the core problem of charging dependency.

OUR VISION

A future where your scooter charges by the sun, so you can go the extra mile.

OUR MISSION

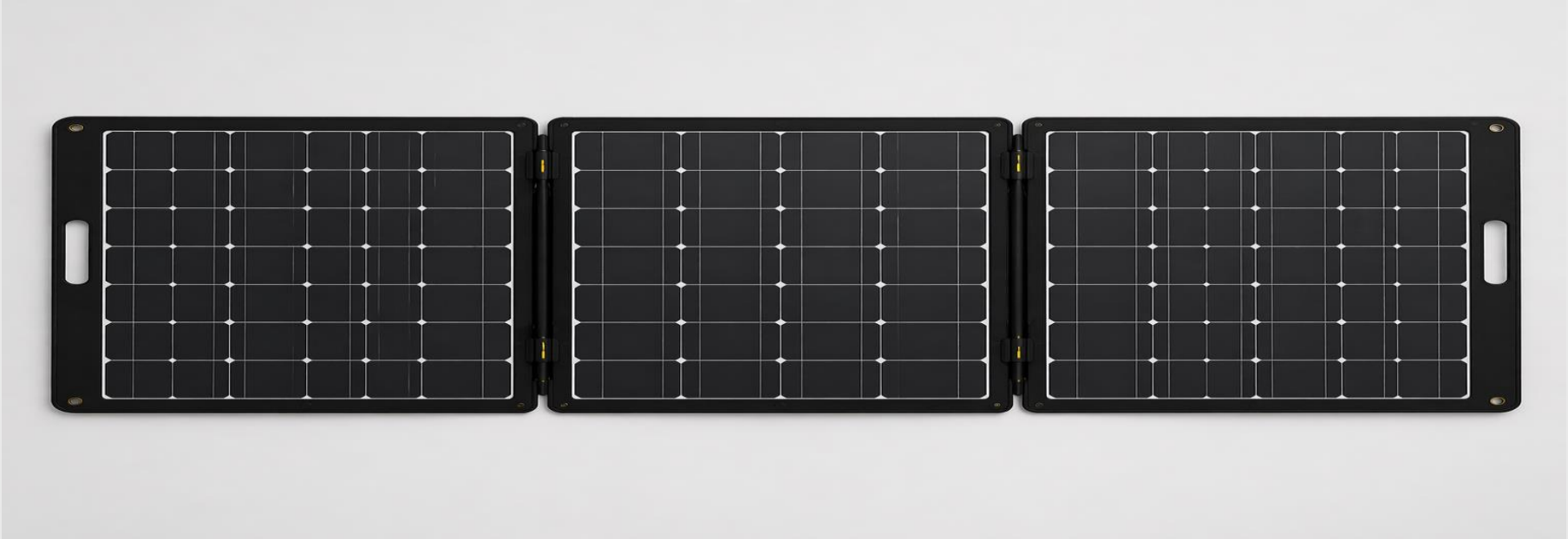
To make EVs more sustainable, self-reliant, and accessible for every Indian.

> "ALL CURRENT SCOOTERS NEED ELECTRICITY TO SURVIVE – OURS CAN GENERATE ITS OWN."

PROPOSED SOLUTION

To address the challenges of limited charging infrastructure and range anxiety, ***we propose the Solar Sync E-Scooter***

An innovative electric scooter equipped with **‘integrated foldable solar panels’** that enable supplementary charging using solar energy.



Integrated Foldable Solar Panels

This dual-charging system allows users to charge their vehicle both:

1. Through conventional plug-in methods, and by
2. Harnessing sunlight when parked outdoors.

The solution **reduces dependency on grid electricity**, **provides additional battery support** during daily use, promotes **Sustainable Mobility** and use of **Green Energy**.

By combining renewable energy with practical design, customized for Indian conditions, this product **enhances convenience**, **improves energy efficiency**, **and makes electric transportation more reliable** and **eco-friendly**.

WE INTRODUCE OUR PRODUCT

SOLAR SYNC ARK α

ARK α

Features:

- ✓ 80 km Range
- ✓ 2.9 kWh Battery Pack
- ✓ 100 W of Solar Panel x3
- ✓ 0.4 – 0.6 kWh Solar Output
- ✓ 10-18 km of Extra Range/day
- ✓ 10–15% extra range
- ✓ Dual Charging System
- ✓ Lithium ion Battery

TARGET AUDIENCE & USE CASE

Target Audience

1. Students & Teacher: Daily travel to school/college limited budget, need cost-saving transport.

2. Office Commuters: Regular city travel need reliable and low-maintenance vehicle.

3. Eco-Conscious Users: People who prefer sustainable lifestyle.

4. Tier-2 & Tier-3 City Users: Limited charging infrastructure.

Use Case

Case 1: College Student

Parks scooter in open parking solar panels charge while he is in class Gains extra 5 – 10% battery.

Case 2: Office Worker

Parks vehicle in sunlight for 6–8 hours Scooter partially recharges automatically.

Case 5: Areas with Power Cuts

No electricity available Solar provides backup charging.

Our product is designed for everyday users who need affordable, reliable, and self-sustaining mobility.

Business Model

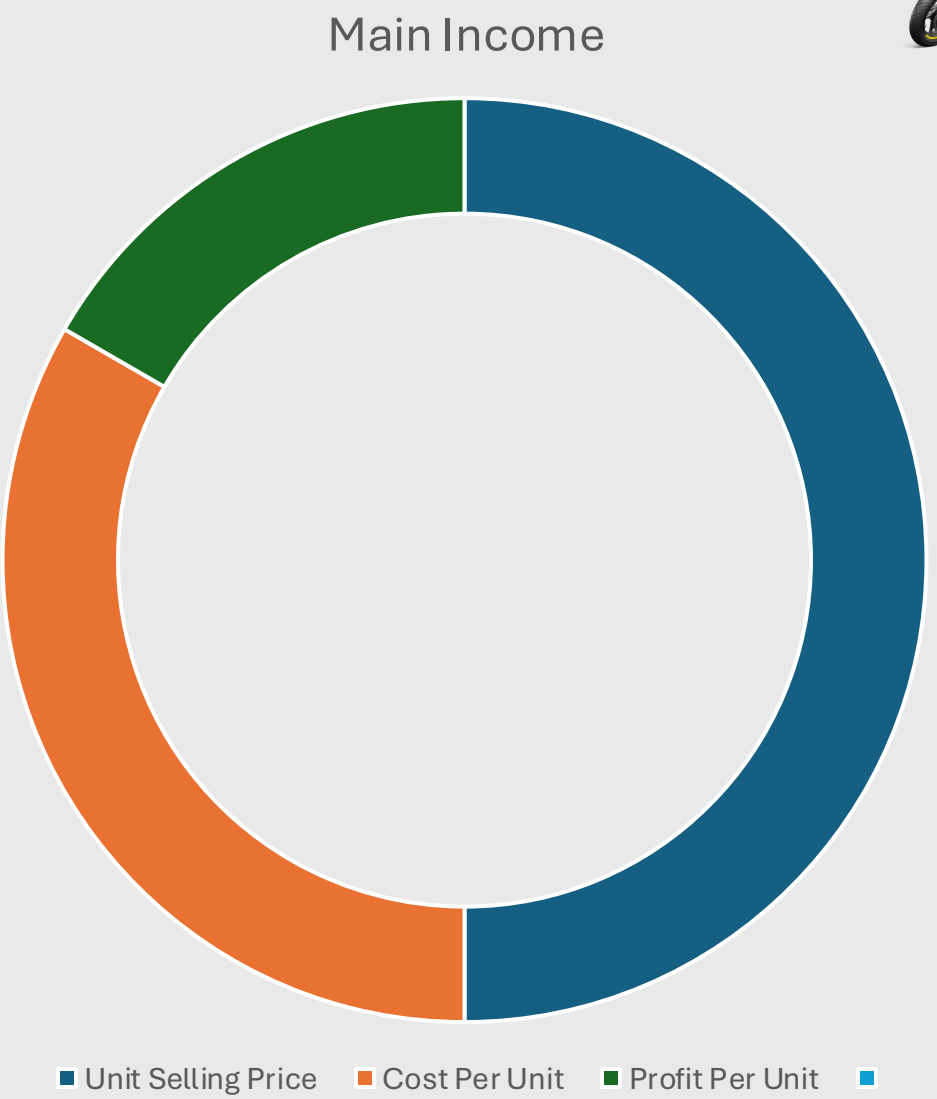
Revenue Streams

Scooter Sales → Main income

Unit Selling Price: ₹95,000

Cost per Unit: ₹80,000

Profit per Scooter: ₹15,000



Other Profits

- **Smart App Subscription:**

Battery analytics, solar tracking, theft alerts.

Price: ₹299/month

- **Accessories & Upgrades:**

Extra solar panels, Fast chargers & Smart add-ons.

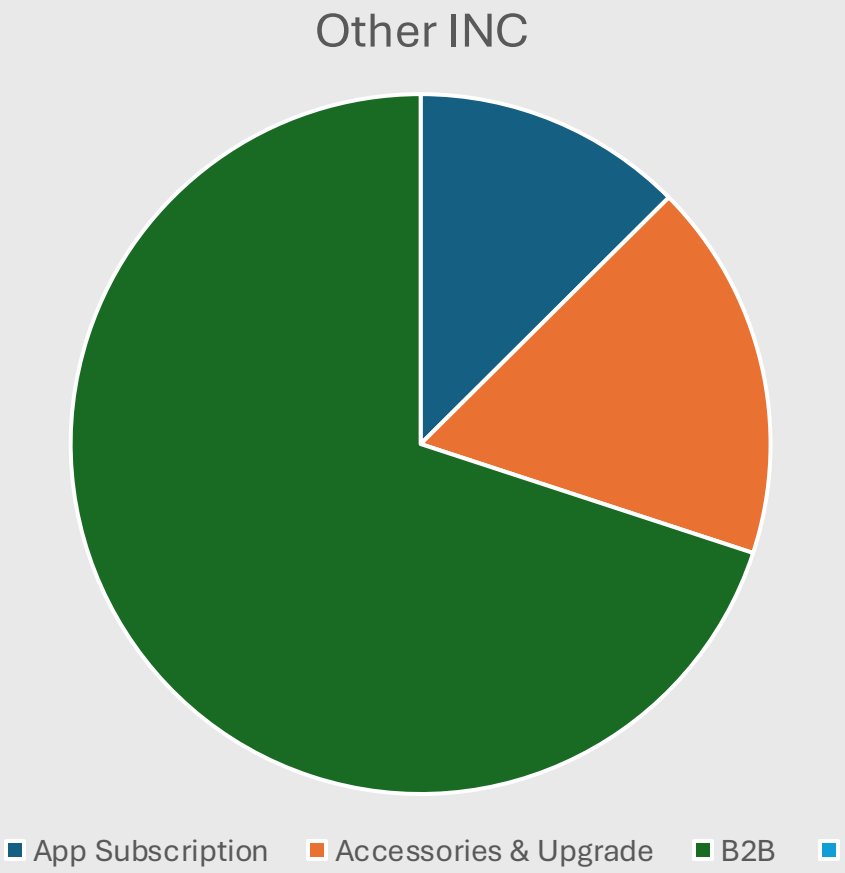
Average revenue per user: ₹2,000 – ₹5,000

- **B2B Partnerships**

Cost/Unit = 80,000

Selling Price = 1,00,000

Profit/Unit = 20,000



TECHNOLOGY USED

1. **Electric Powertrain: Lithium-ion Battery** (high efficiency, lightweight).

2. **Solar Charging System**: **Foldable Photovoltaic (PV) Panels** integrated into scooter body. Converts sunlight into electrical energy. Provides **supplementary charging (5–10% boost/day)**.

3. **Dual Charging System:** Standard **AC Plug Charging** (home/charging station) **Solar-assisted charging** (when parked outdoors).

4. **Battery Management System (BMS):** Monitors battery health & temperature

Prevents: Overcharging & Overheating

Ensures safety & longer battery life

We are not just building a scooter, we are building a smart energy system on wheels.

Dual Charging System – Plug in at home or charge with sunlight when outdoors.

More power. More freedom.

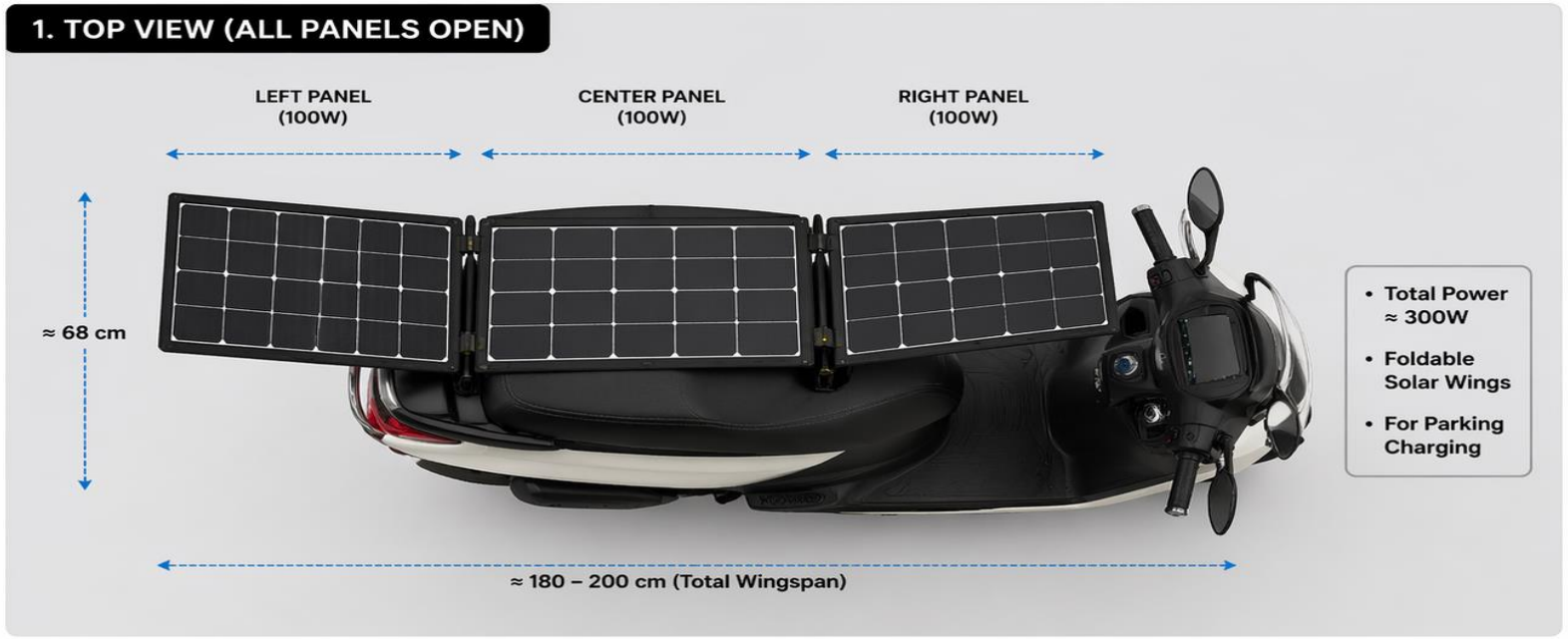
INDIAN ADAPTATION

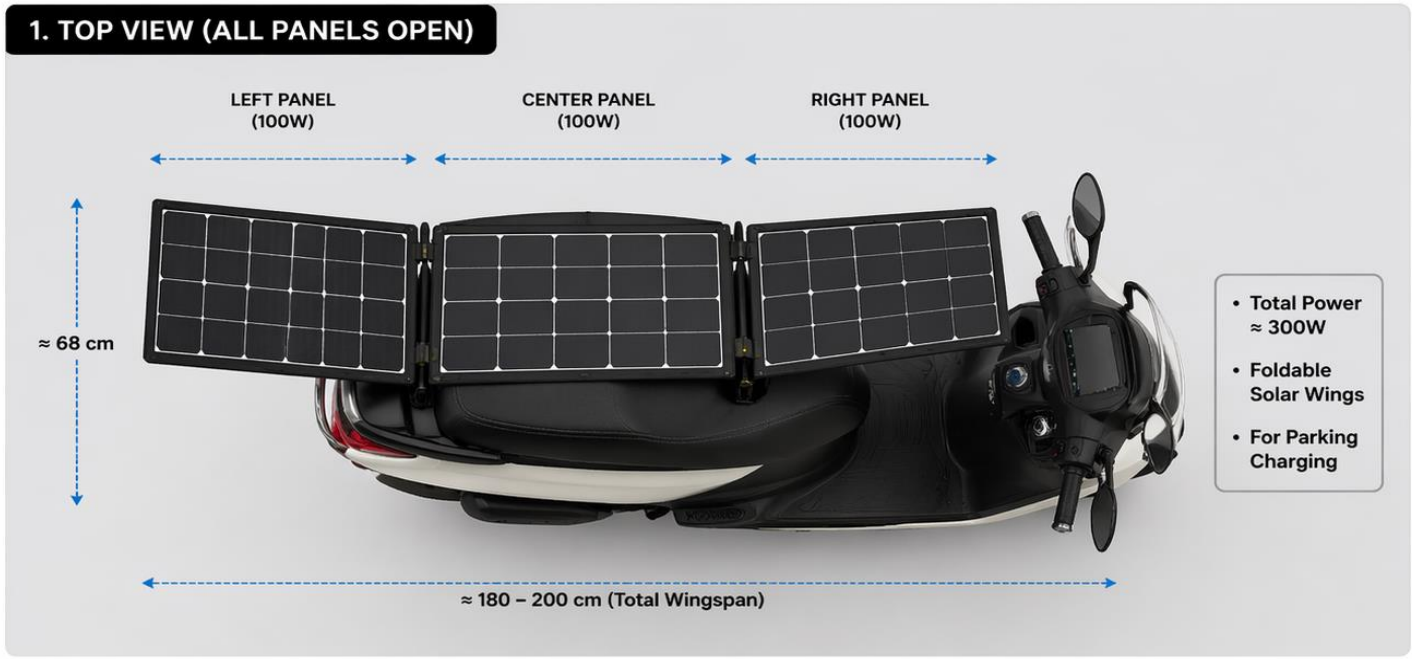
Why it works in India?

India has high sunlight availability and still many vehicles are parked outdoors direct under the sun, E.V. demand is rapidly growing where as charging infrastructure is still limited.
Useful in rural areas (less charging stations)Supports government EV initiatives and reduces electricity dependency.

Ark $\boldsymbol{\alpha}$ is an Indian solution for a Indian Problem.

PROTOTYPE / SAMPLE




1. TOP VIEW (ALL PANELS OPEN)
LEFT PANEL (100W)
CENTER PANEL (100W)
RIGHT PANEL (100W)
≈ 68 cm
≈ 180 – 200 cm (Total Wingspan)
• Total Power ≈ 300W
• Foldable Solar Wings
• For Parking Charging

2. LEFT CORNER TOP VIEW
LEFT PANEL OPENS OUTWARD

3. RIGHT CORNER TOP VIEW
RIGHT PANEL OPENS OUTWARD

IMPACT & BENEFITS

1. Reduces dependency on electricity from fossil fuels
2. Promotes clean, renewable solar energy
3. Lowers carbon emissions in daily transport
4. Saves electricity cost over time
5. Extra **5–10% battery backup daily**
6. Encourages adoption of Ev(s)
7. Supports India’s sustainable mobility goals

Our solution is not just eco-friendly, but also cost-effective and practical for everyday Indian users.

FEASIBILITY & SCALABILITY

- Uses **existing technologies** (EV + Solar panels)
- No need for new infrastructure
- Can be integrated into current scooter designs
- Solar panels are already widely available
- Manufacturing can be done using existing EV supply chains

Scalability

Can expand across:

1. Urban cities
2. Tier-2 & Tier-3 towns
3. Rural areas

INNOVATION & UNIQUENESS

What Makes It Unique?

1. Dual Charging System
2. Self-charging capability using sunlight

Innovation Angle!

- ✓ Combines **renewable energy + mobility**
- ✓ Converts idle parking time into energy generation
- ✓ Reduces dependency on infrastructure

While others improve batteries, we change the way energy is generated!

THANK YOU

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